

|   |                   |                                            |      |  |    |     |
|---|-------------------|--------------------------------------------|------|--|----|-----|
| 7 | (a) <i>either</i> | $V = IP$                                   |      |  | B1 |     |
|   |                   | current in circuit = $E / (P + Q)$         |      |  | B1 |     |
|   |                   | hence $V = EP / (P + Q)$                   |      |  | A0 | [2] |
|   | <i>or</i>         | current is the same throughout the circuit | (M1) |  |    |     |
|   |                   | $V / P = E / (P + Q)$                      | (A1) |  |    |     |
|   |                   | hence $V = EP / (P + Q)$                   | (A0) |  |    |     |

| Page 4 | Mark Scheme                            | Syllabus | Paper |
|--------|----------------------------------------|----------|-------|
|        | GCE A/AS LEVEL – October/November 2008 | 9702     | 02    |

- (b) (i) (as temperature rises), resistance of (thermistor) decreases M1  
*either* resistance of parallel combination decreases  
*or* p.d. across 5 k $\Omega$  resistor / thermistor decreases M1  
p.d. across 2000  $\Omega$  resistor / voltmeter reading increases A1 [3]
- (ii) if  $R$  is the resistance of the parallel combination,  
*either*  $3.6 = (2 \times 6) / (2 + R)$  *or* current in 2 k $\Omega$  resistor = 1.8 mA C1  
 $R = 1.33 \text{ k}\Omega$  current in 5 k $\Omega$  resistor = 0.48 mA C1  
 $\frac{1}{1.33} = \frac{1}{5} + \frac{1}{T}$  current in thermistor = 1.32 mA C1  
 $T = 1.82 \text{ k}\Omega$   $T = 2.4 / 1.32 = 1.82 \text{ k}\Omega$  A1 [4]